The split verdict

What AI actually fixes in engine health monitoring — and what only sensors can

*One-page summary of **EHMbrAIn**: AI-based vs. traditional EHM, a pre-registered gas-path-analysis testbed on the CFM56-7B with synthetic ground truth*

The question, made falsifiable

Does AI beat the traditional engine-health-monitoring stack — or is that marketing? We built the comparison no field study can run: a pyCycle thermodynamic twin of the CFM56-7B26 calibrated to certificate data, a 100-engine synthetic fleet (SynCFM56) with known ground-truth health, and two *tuned* practitioners — the classical stack (Kalman GPA, CUSUM, trending) and an AI suite (Mahalanobis, GRU, conformal) — given identical data, identical sensors, and a symmetric 50-trial tuning budget. Every hypothesis, threshold, and test was frozen in twelve tagged pre-registrations *before* the test set was touched; failed gates were repaired physically, never by moving thresholds.

Verdicts (single confirmatory pass, Holm-corrected)

Task	Trad.	AI	Verdict
Detection recall (FA 1=1)	0.13	**0.48**	AI, p=.008
median delay (cycles)	6033	**499**	12× earlier
Isolation, confusable faults	31 %	31 %	*tie: wall*
RUL RMSE @90 % life (cy)	1981	**858**	AI, δ=0.89
vs. advanced classical	1118	**858**	AI, p=.002
90 % interval half-width (cy)	11509	**1957**	5.9× tighter
Physics-informed hybrid	—	—	refuted 2×

The pattern survives contact with reality: on NASA's C-MAPSS benchmark the same pipelines score 14.9 vs. 42.1 cycles RMSE, and the classical side was even upgraded mid-study to a similarity-based prognostic (1981→1118) — AI still wins (858, p=0.002).

The split is physics, not fashion

AI wins exactly where the task is *statistical aggregation over time*: pulling faint trends out of noise (detection), integrating a whole degradation history into a forecast (prognosis), and honestly sizing its own error (uncertainty). It ties exactly where the *sensor set* cannot distinguish two faults: the cockpit suite's influence-matrix geometry leaves seven confusable fault pairs at ∼1.3° separation, and no model — classical or learned — can invert information that never reached the data. Proof: adding two real station sensors lifts confusable isolation from 0.15 to 0.92 (p=0.001); adding *virtual* (model-derived) versions of the same sensors changes nothing (p=1.0). Where learning does add isolation power is fusion: a GRU fusing multi-flight physics projections more than triples classical multi-point analysis (0.65 vs. 0.17) on identical windows, and the ACARS reporting *schedule* itself becomes a design variable (+79 % separability from one extra stabilized-cruise report).

Certificates, not promises

Two by-products may outlast the scoreboard. An *identifiability certificate* — accumulated Fisher information over an engine's actual flight history — predicts per-direction diagnosis error *before* deployment, validated against ground truth (Spearman ρ=0.70): it tells an operator which faults their fleet can see and which sensor rescues which blind spot. A *prognostic floor* — the aleatoric RUL uncertainty that no model can remove — shows 87 % of early-life error is irreducible, while at 90 % of life the best method still sits 4× above the floor: buy algorithms for the endgame, calibration for the rest.

What it is worth

The operational currency is converting unscheduled removals into scheduled ones. Against the floor-derived ceiling, the AI stack converts a net 35–50 % of unscheduled events across realistic lead-time demands; the traditional stack manages 0–25 %. For a 50-aircraft, 100-engine fleet, a Monte-Carlo cost model prices the difference at $3.2M/yr median ($0.9M–$8.0M p10–p90, <1 % probability of net loss).

Takeaway: "AI vs. traditional" is the wrong question. AI is a strictly better *statistician* on the same data; it is not a new *sensor*. Buy AI for detection, prognosis, and calibrated uncertainty; buy instrumentation for isolation; and demand certificates — identifiability and floor — that say, in advance, which regime you are in.

Full study: 106-page report, 12 pre-registration tags, every figure and number auto-generated from versioned artifacts, replicable end-to-end on a desktop (`make all`, ∼11 min). Repository: github.com/cedarcreeks/EHMbrAIn.